

Ethanol Blending in Gasoline – Malta

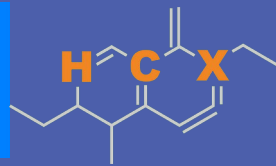
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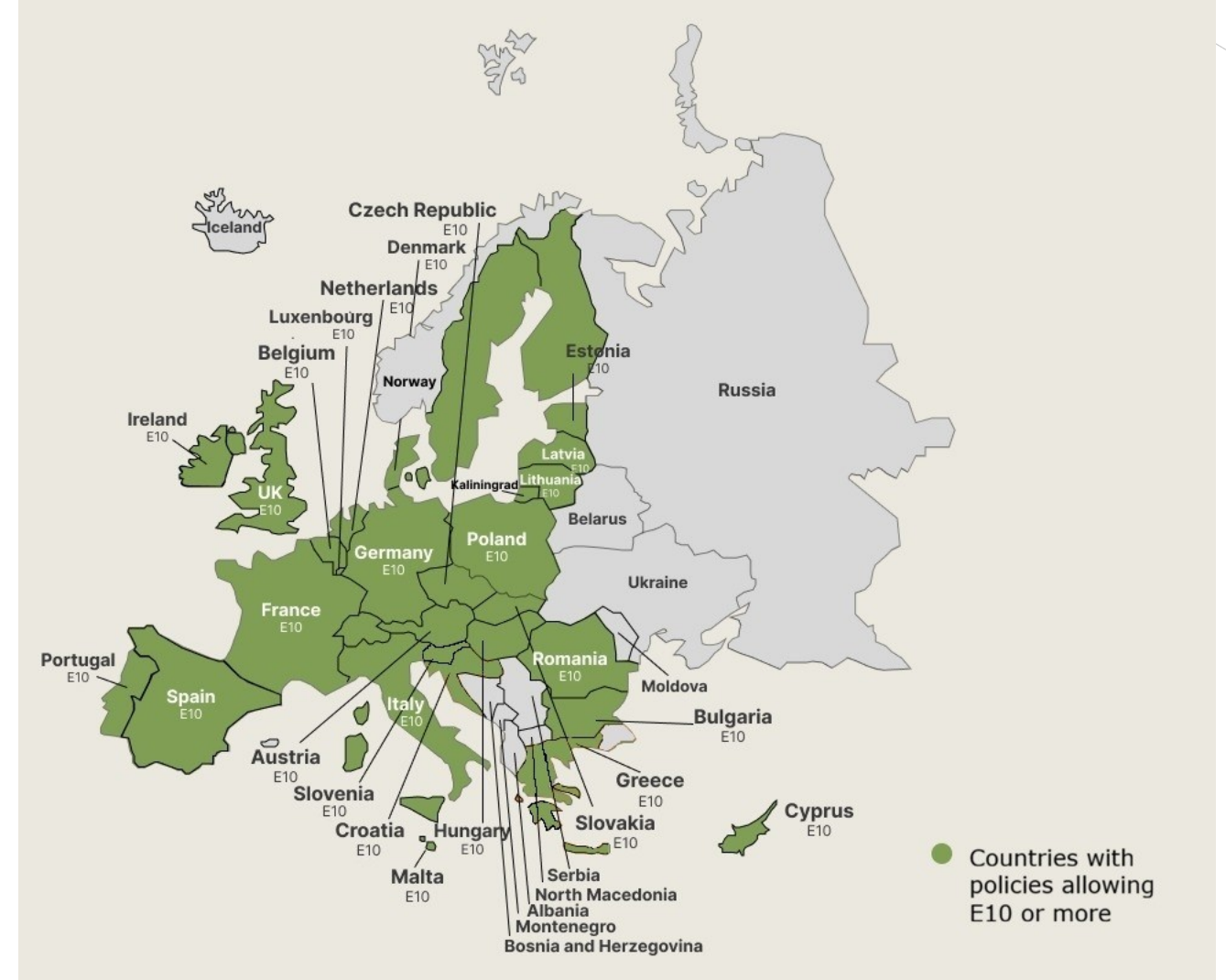


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Ethanol Blending in Europe

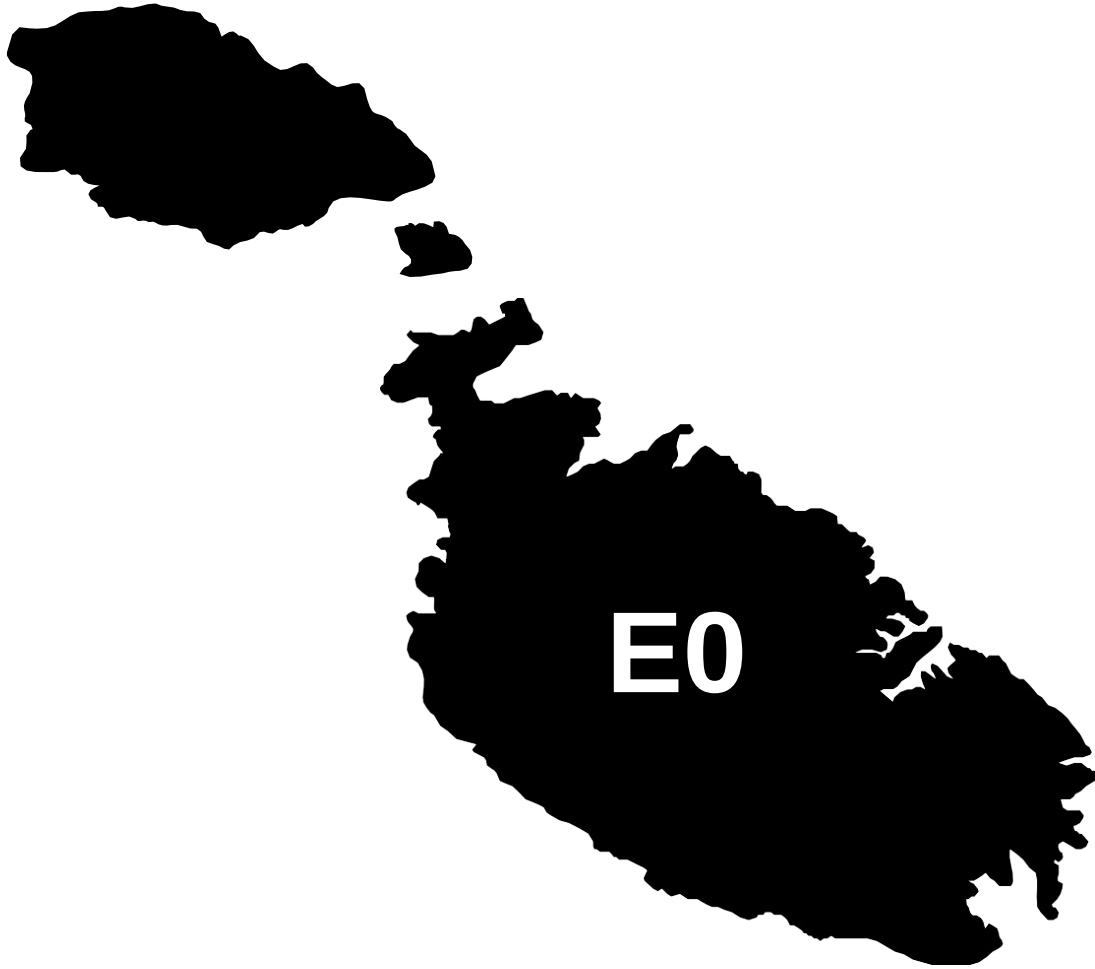
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

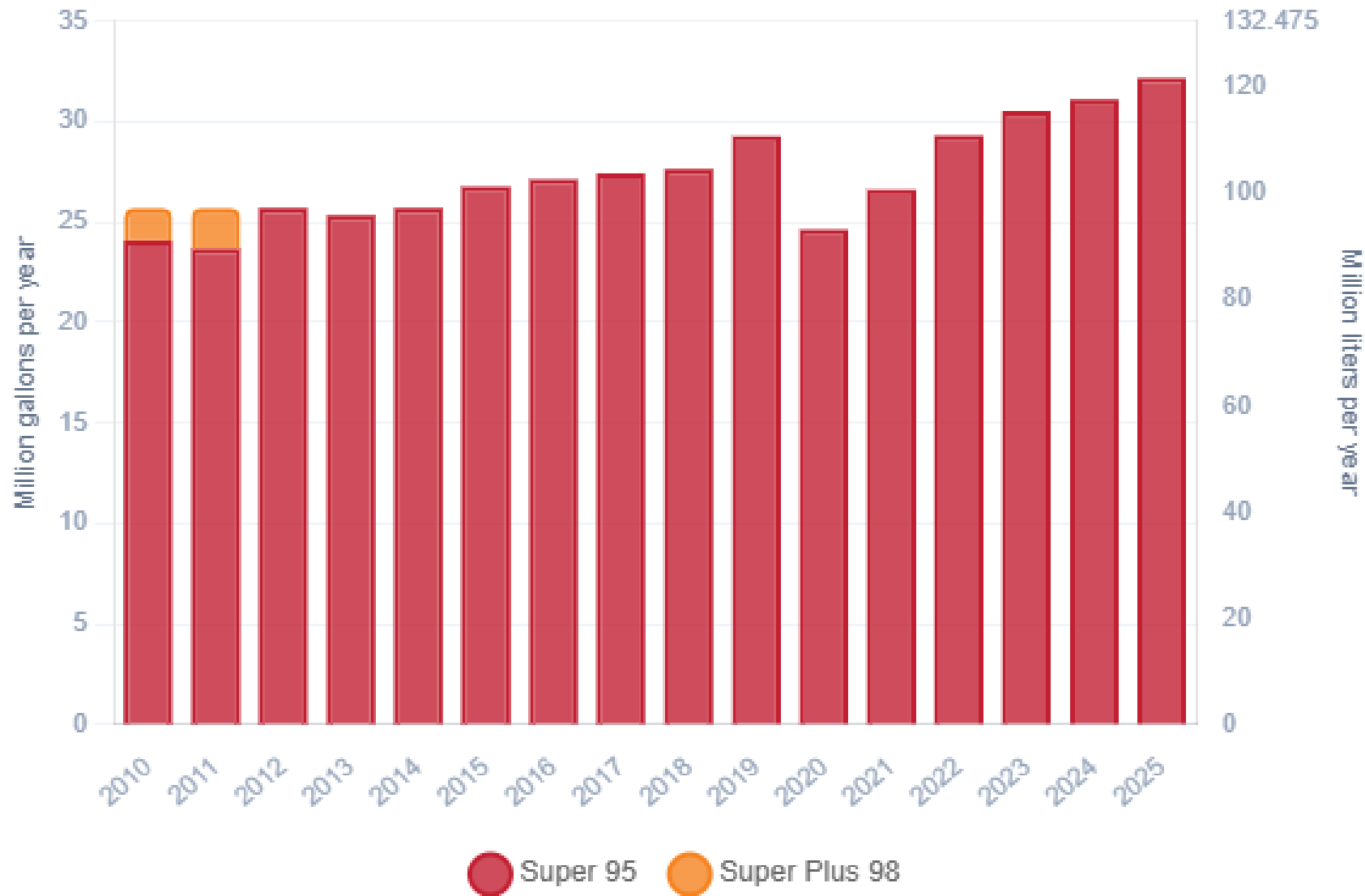
Ethanol Gasoline Blending in Malta



In 2025, gasoline consumption in Malta was 122 million liters (32 million gallons) and growth rate was 1.5%. There is no gasoline production. Gasoline imports were 74 million liters (20 million gallons) . Malta complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.0 RON.

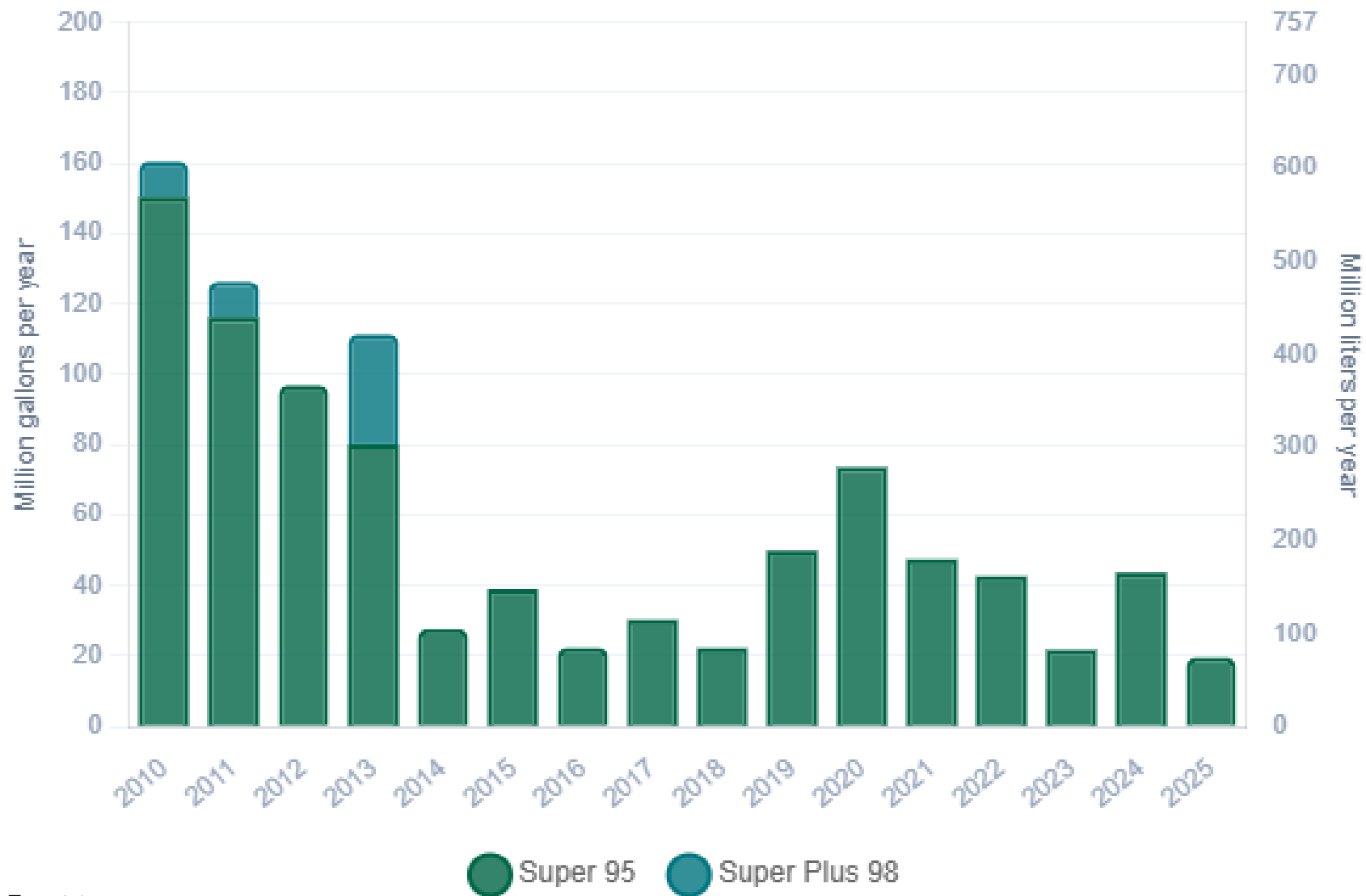
E10 is allowed but Malta does not produce or consume bioethanol at a commercial level

Gasoline Demand in Malta



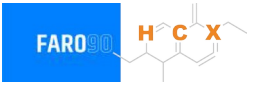
Source: Eurostat

Gasoline Imports in Malta



Source: Eurostat

Ethanol Balance in Malta



Malta does not produce or consume bioethanol at a commercial level

Source: Eurostat

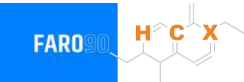
Gasoline Quality in Malta

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Countries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.2							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commission

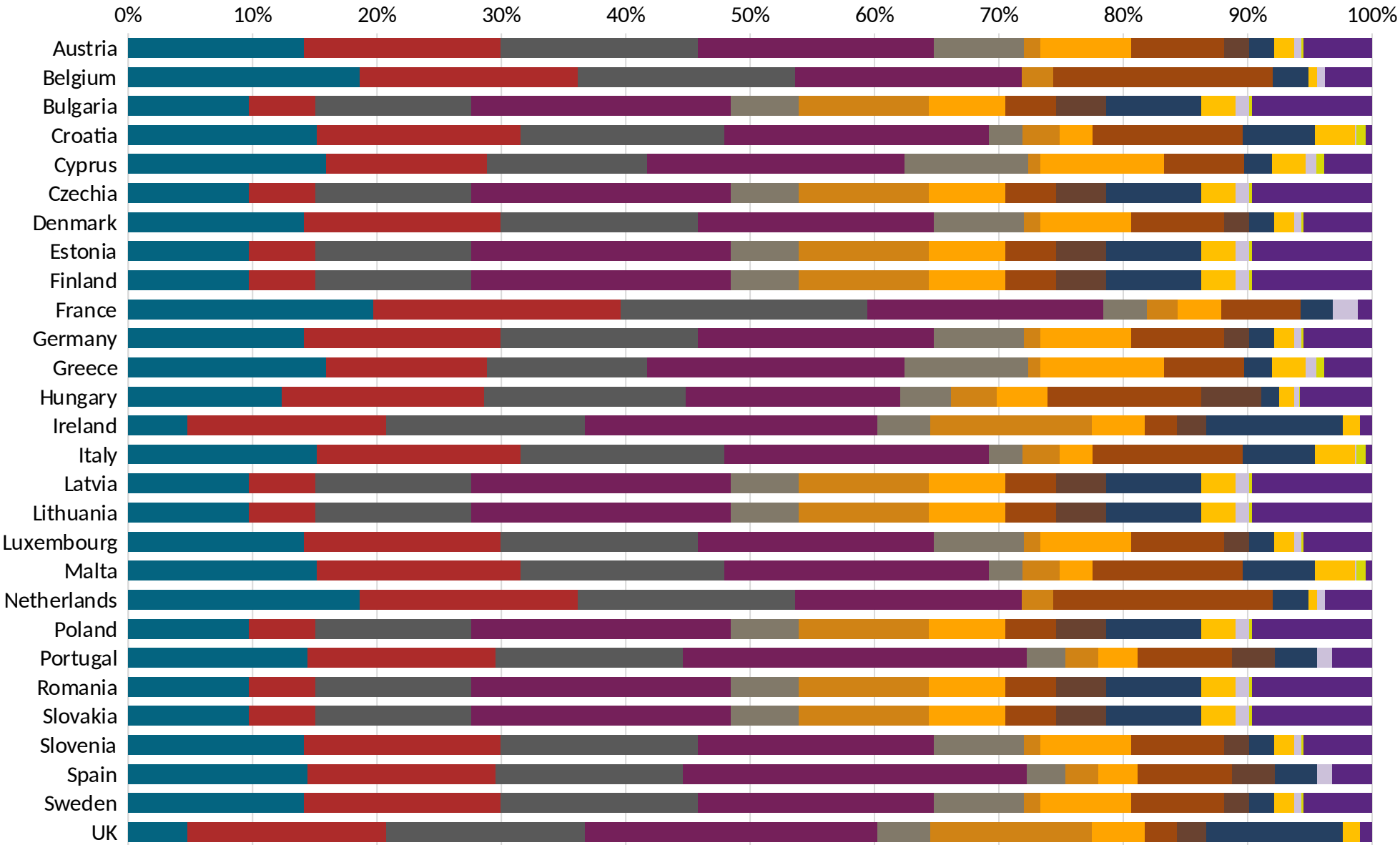
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source:
Faro90



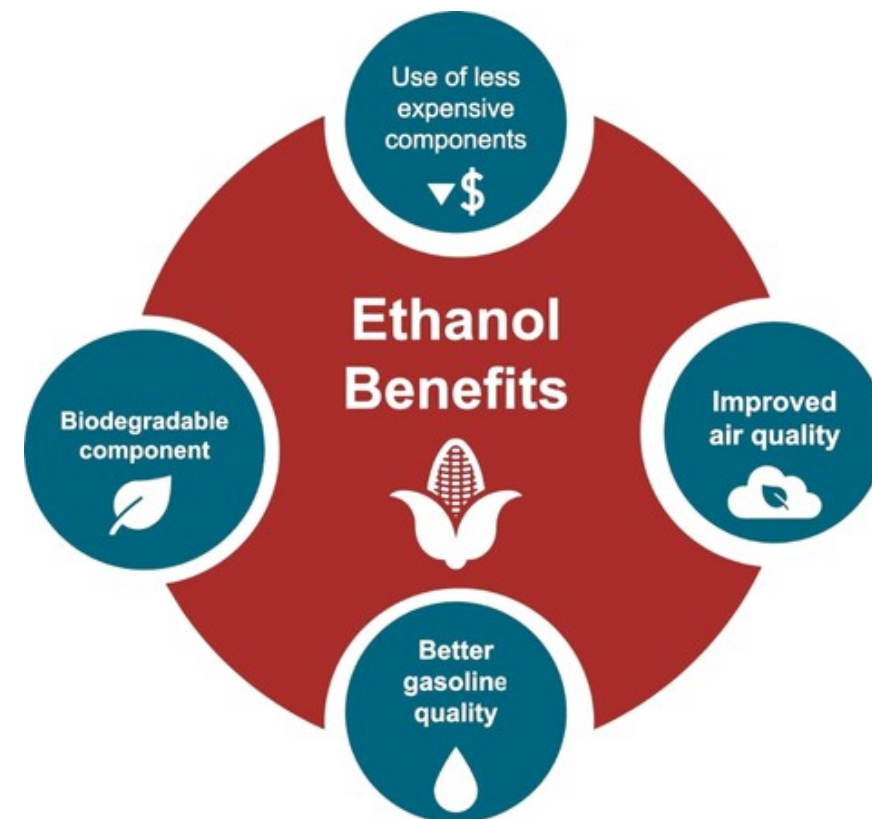
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

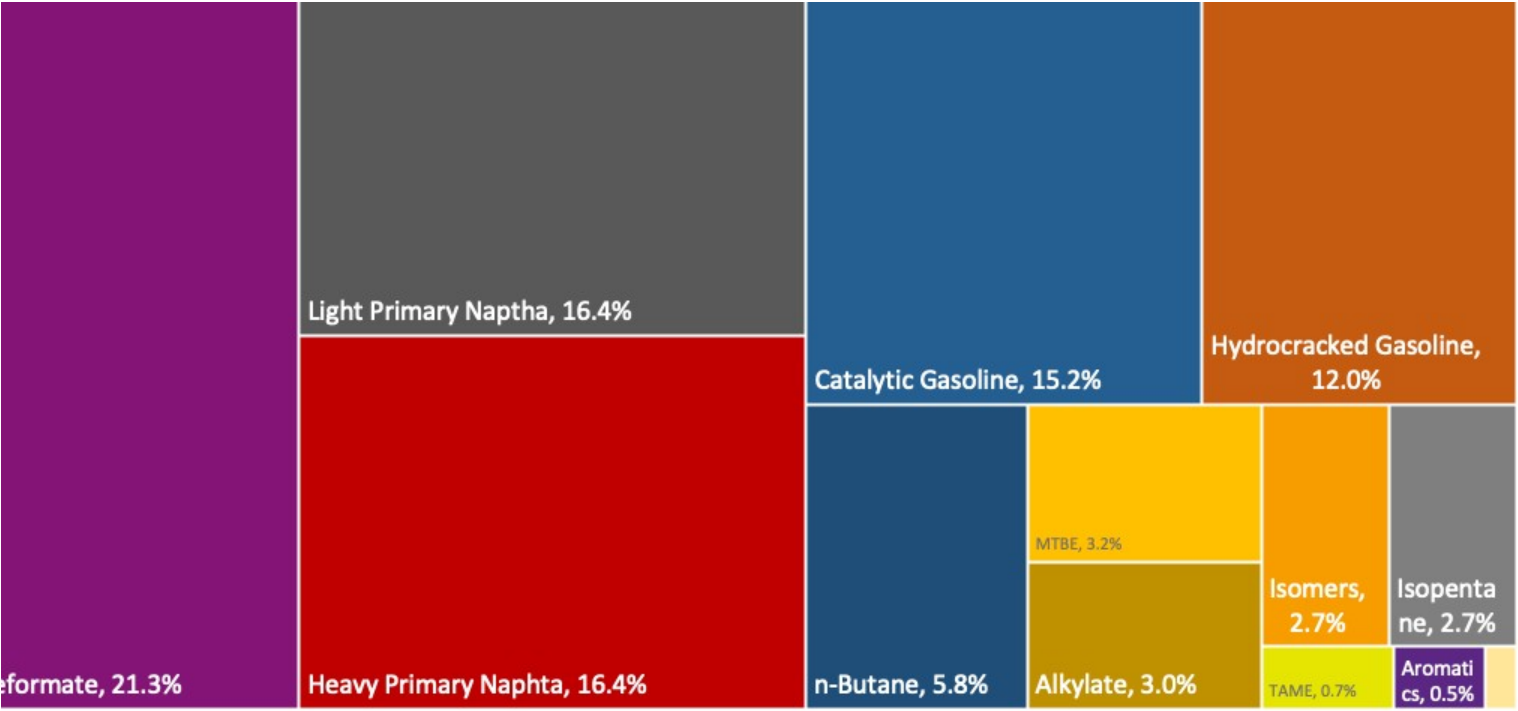
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

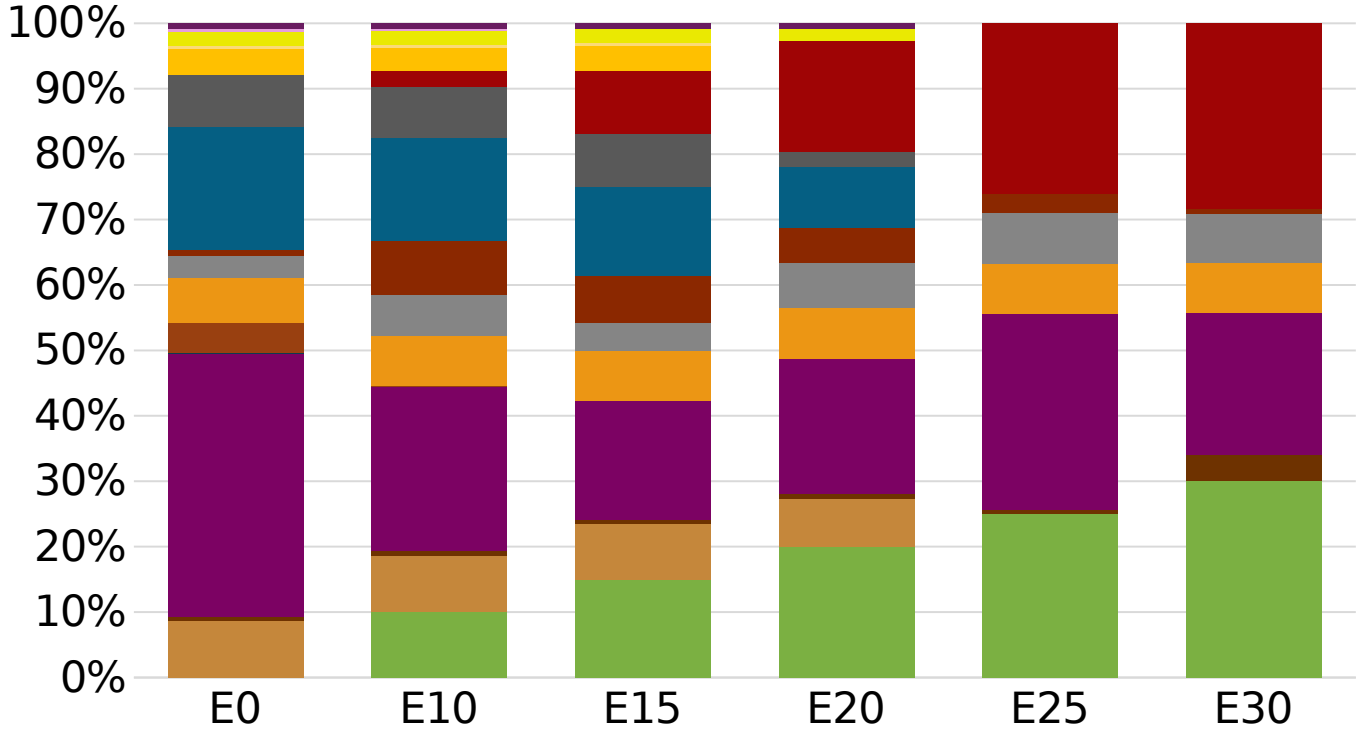
The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Malta Available Blending Components



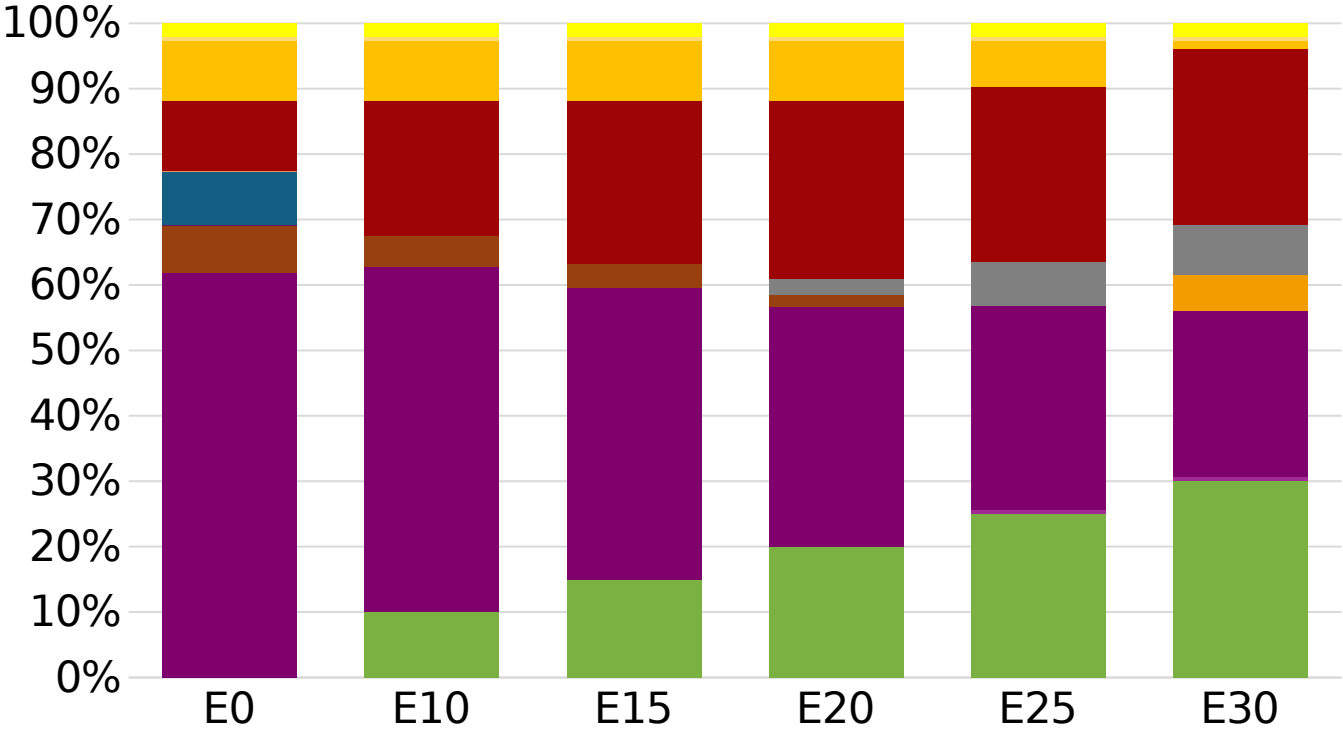
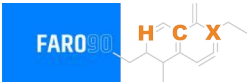
Malta – Gasoline 95 – Constant Octane



Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US\$/gal)	\$ 2.055	\$ 1.942	\$ 1.859	\$ 1.783	\$ 1.688	\$ 1.614

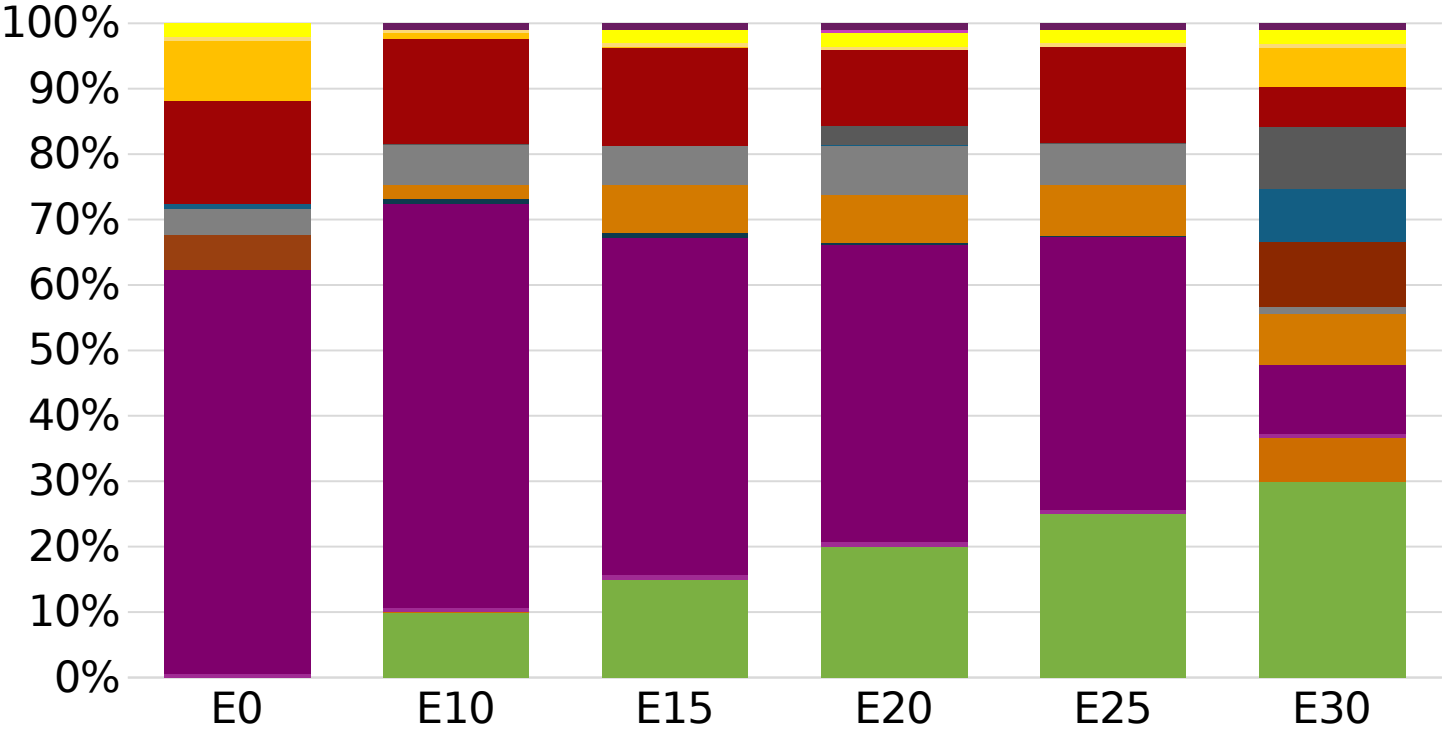
Malta - Gasoline 98 - Constant Octane



Average CIF 2025
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.104	\$ 1.985	\$ 1.916	\$ 1.846	\$ 1.776	\$ 1.723

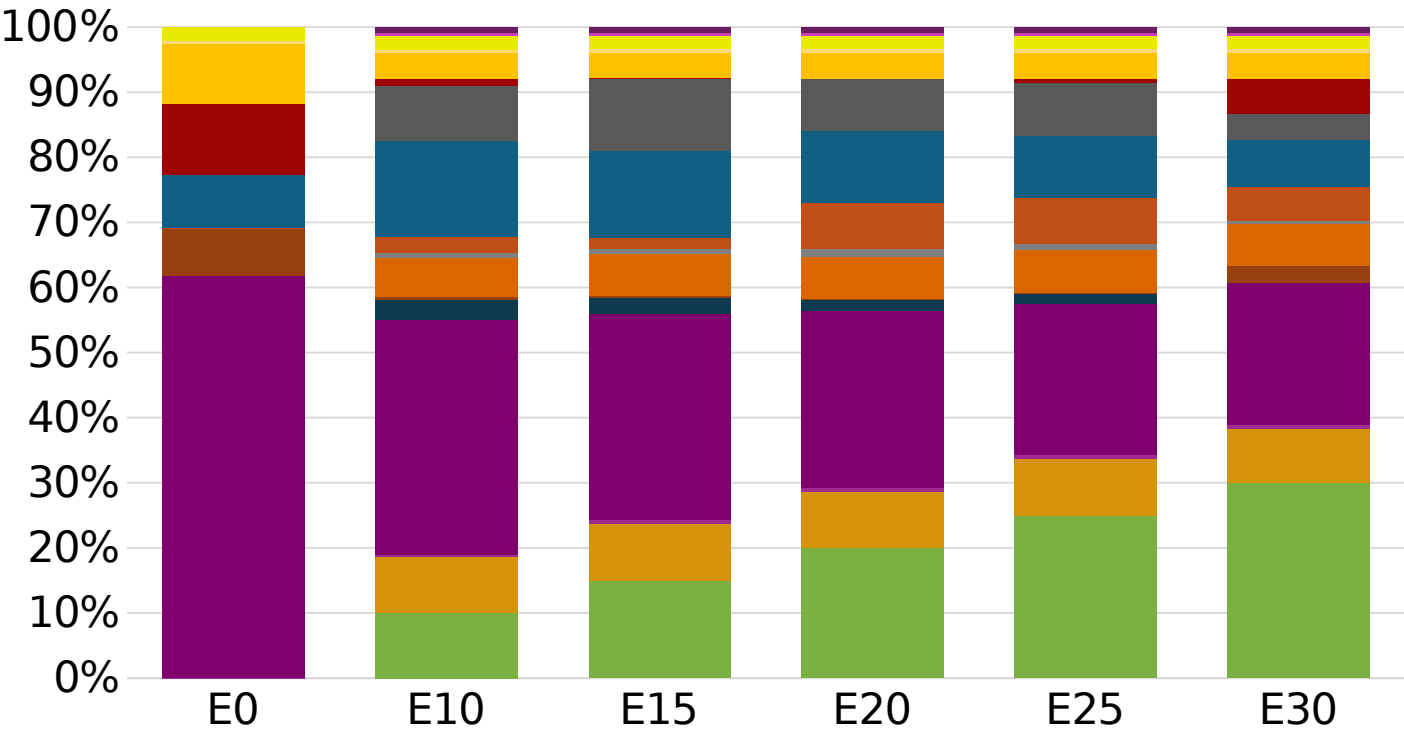
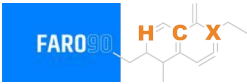
Malta – Gasoline 95 – Increased Octane



Average CIF 2025 Prices

Octane (RON)	95.0	95.9	97.5	99.0	100.5	101.8
Price (US\$/gal)	\$ 1.974	\$ 1.974	\$ 1.974	\$ 1.974	\$ 1.974	\$ 1.974

Malta – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	98.0	98.8	100.1	101.3	102.6	104.1
Price (US\$/gal)	\$ 2.104	\$ 2.104	\$ 2.104	\$ 2.104	\$ 2.104	\$ 2.104

Source:
Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

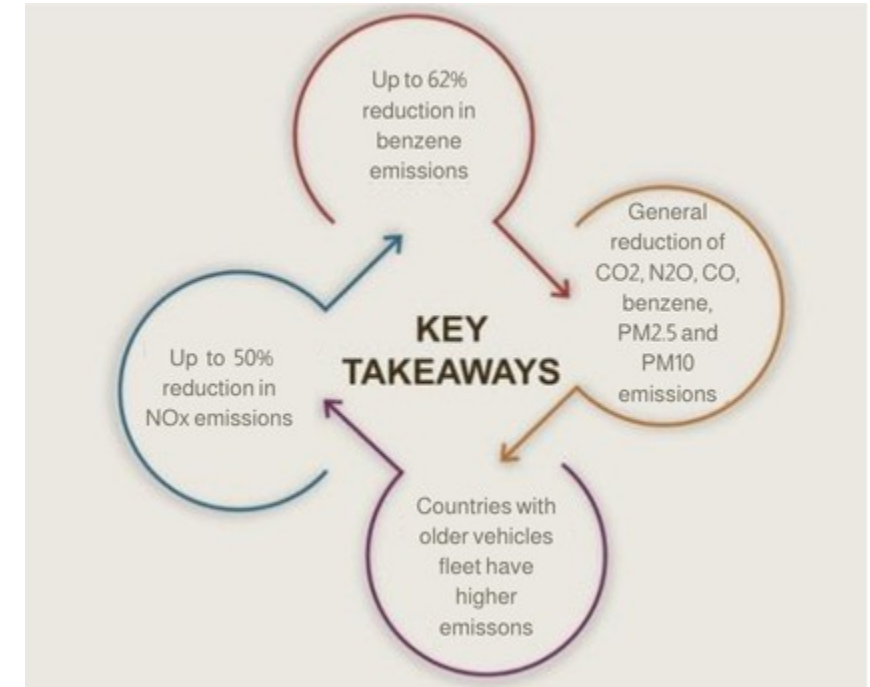
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

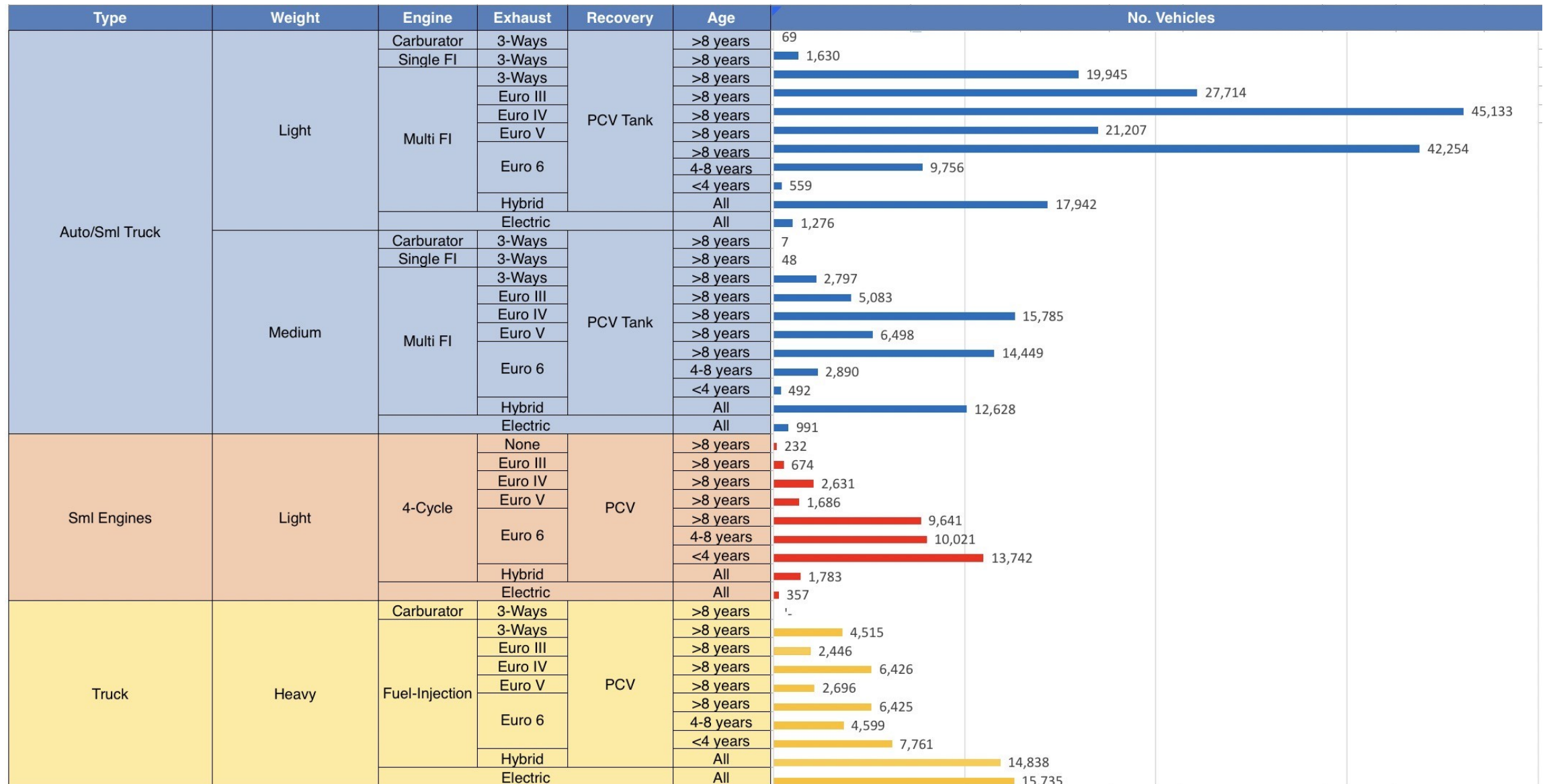
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - Malta

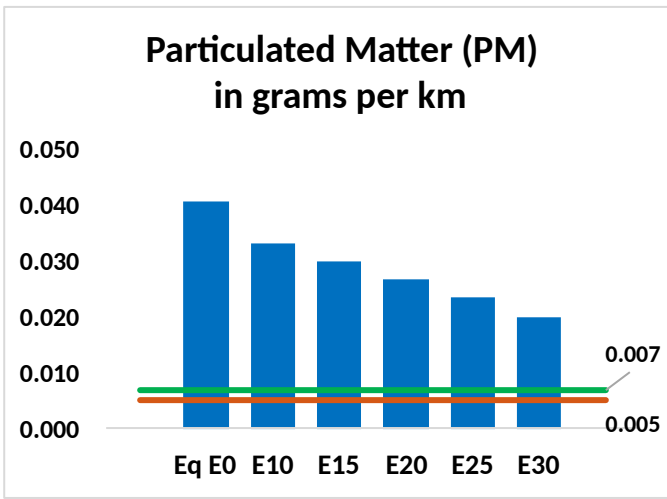
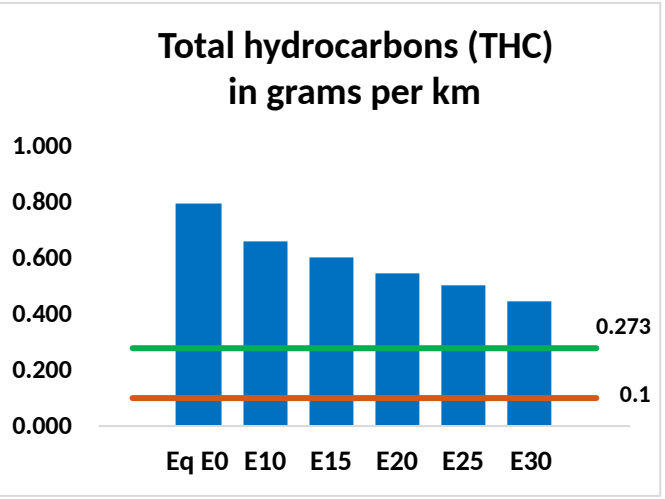
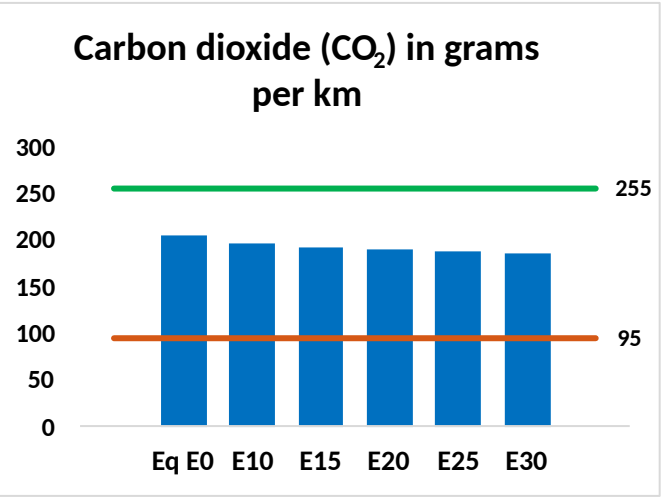
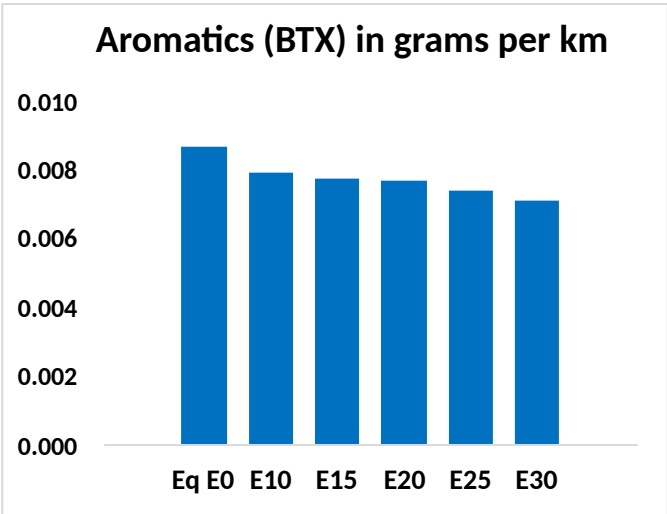
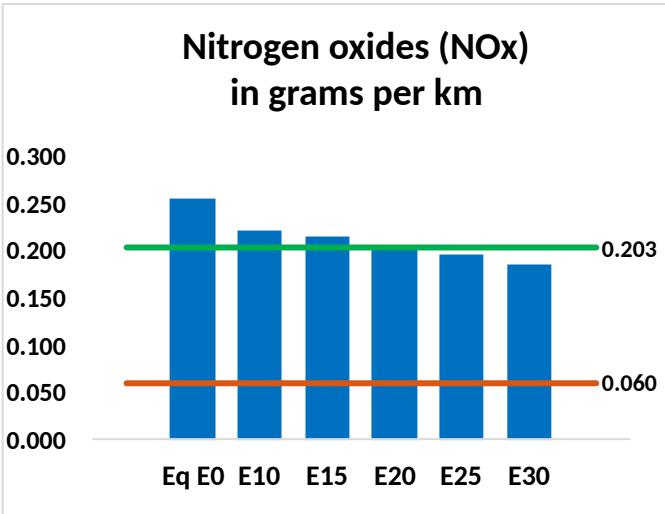
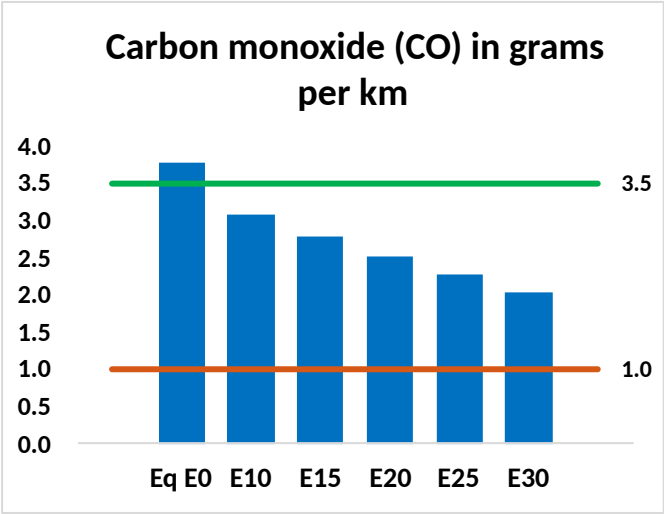
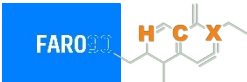


Vehicle Fleet: 355,360
Gasoline Vehicle Fleet 200,048

Source: Eurostat, ACEA

Malta – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Malta – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	18,330	16,673	15,017	13,612	12,175	11,064	9,890	-18%	-34%
CO2	1,003,265	977,467	953,102	933,940	924,458	919,120	902,178	-5%	-8%
VOC	3,880	3,550	3,220	2,943	2,662	2,445	2,158	-17%	-31%
NOx	1,243	1,148	1,083	1,049	996	954	904	-13%	-20%
SOx	12	11	10	9	8	8	7	-15%	-28%
NH3	388	384	380	382	386	389	392	-2%	0%
N2O	14	14	14	14	15	15	15	-1%	2%
CH4	786	786	786	797	815	830	844	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	19	18	17	17	16	16	15	-9%	-14%
Acetaldehyde	63	73	107	163	222	253	299	68%	249%
Formaldehyde	238	232	270	317	332	367	400	13%	39%
Benzene	42	41	39	38	38	36	35	-9%	-11%
PM2.5	60	55	49	44	40	35	29	-18%	-34%
PM10	138	126	113	102	91	80	68	-18%	-34%

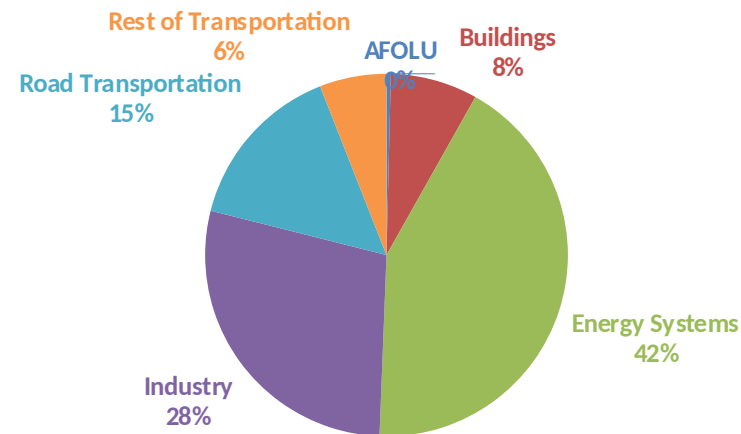
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

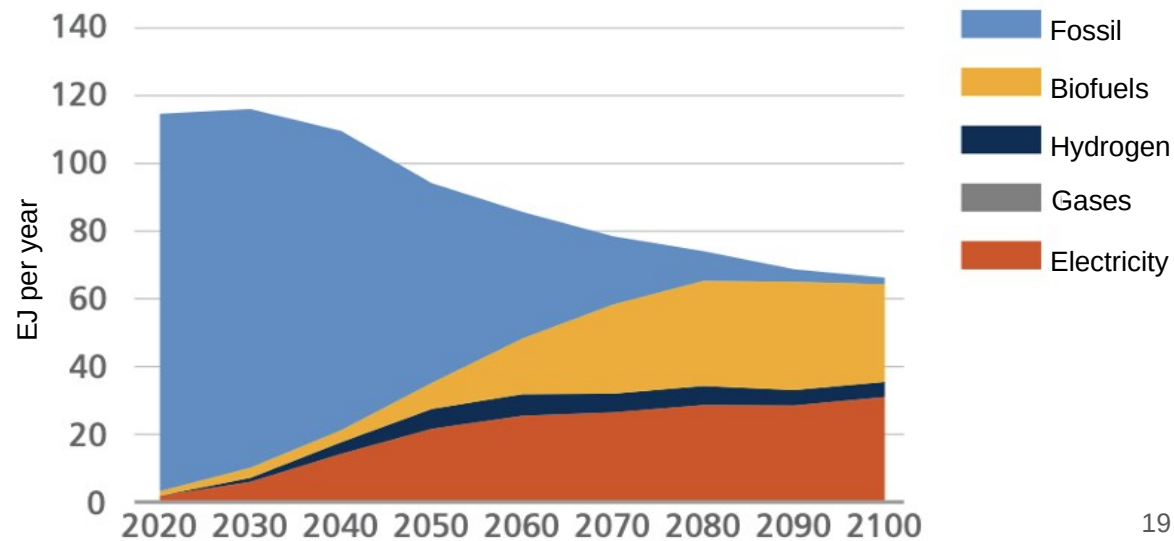
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

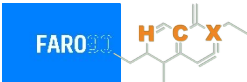
Current GHG Global Emissions Distribution



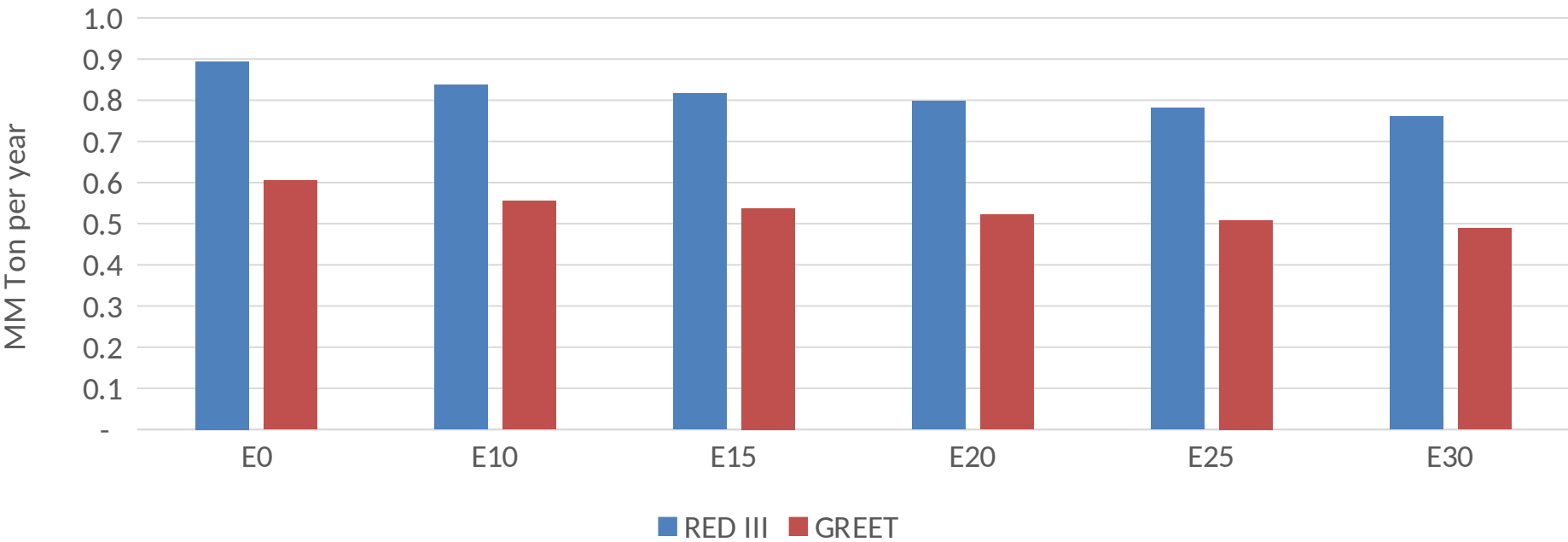
IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Malta – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Life Cycle GHG Emission - Gasoline Vehicles



Country Emissions 2025 (MMT/año)	2.2
Target @ 2035 (MMT/year)	1.2
Delta	1.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.2%	11.1%	13.7%	15.9%	19.1%
RED II/III	0.0%	6.3%	8.6%	10.7%	12.5%	14.9%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.2%	7.0%	8.7%	10.1%	12.0%
RED II/III	0.0%	5.9%	8.0%	10.0%	11.7%	13.9%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90